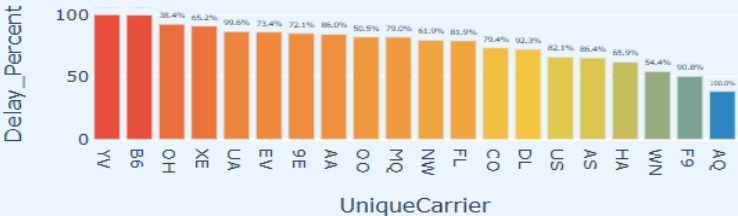


✈️ Aviation Analytics Dashboard – Big Data Capstone

Integrating Flight Delays • Passenger Satisfaction • Route Optimization • Social Sentiment • Pricing Insights

✈️ Case 1 Insight: Airlines with higher delay % need predictive scheduling & maintenance models.

✈️ Case 1: Average Flight Delay by Airline



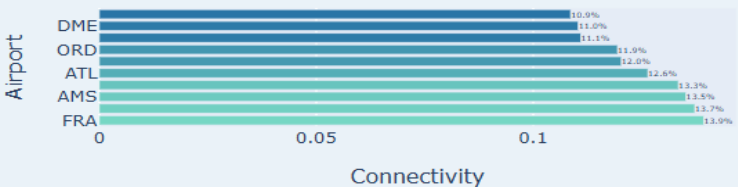
😊 Case 2 Insight: Business class passengers have the highest satisfaction — premium service consistency.

😊 Case 2: Passenger Satisfaction by Class



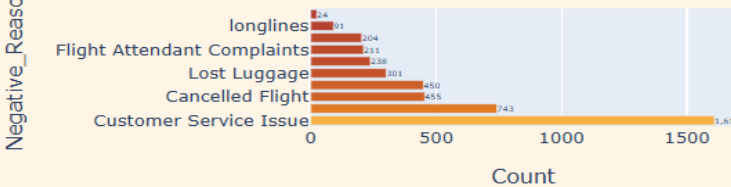
🌐 Case 3 Insight: Airports like LHR, ATL, and DXB act as central hubs — key for route optimization.

🌐 Case 3: Top Connected Airports



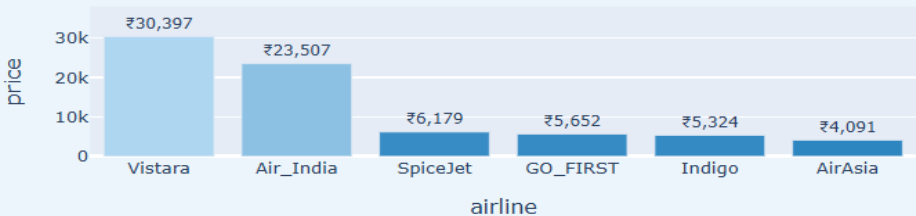
🚨 Case 4 Insight: Most complaints are due to delays — AI-driven sentiment monitoring can help resolve issues.

🚨 Case 4: Top Reasons for Negative Tweets



💰 Case 5 Insight: Business fares are 3× economy — dynamic pricing can maximize profits & manage demand.

💰 Case 5: Average Flight Price by Airline



📊 Key Takeaways

- Predictive models can reduce flight delays.
- Enhance customer experience via personalization.
- Optimize global routes for fuel & time efficiency.
- Sentiment analytics improves brand trust.
- Dynamic pricing ensures better revenue management.